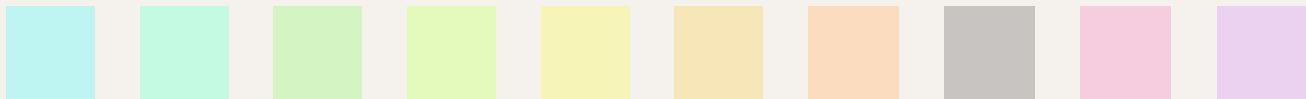


Client Side HTML

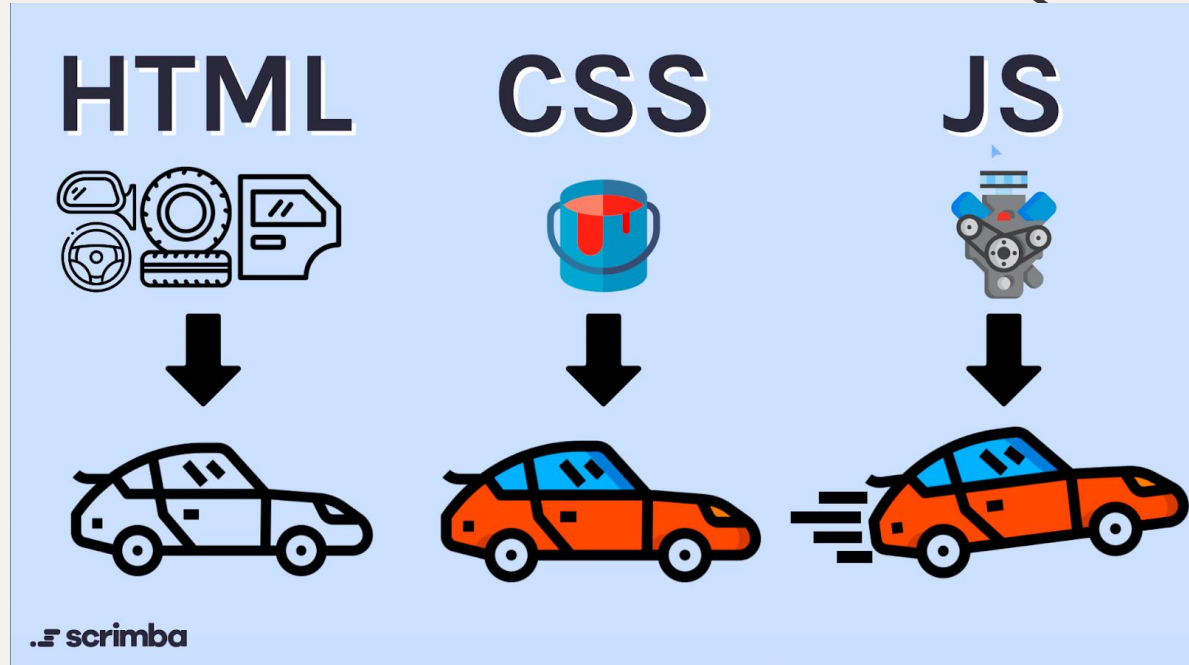
מבוא לבניית מערכות ממוחשבות מבוססות אינטרנט הרצאה 2 – HTML



HTML- Content

CSS- Design

JavaScript- Functionality



<https://www.sbs.com.au/theboat/>

HTML



HTML: a HyperText Markup Language

- **Markup language** - a computer language that defines the structure and presentation of raw text.
- **Hypertext** - text displayed on a computer or device that provides access to other text through links, also known as hyperlinks.
- In HTML, the computer can interpret raw text that is wrapped in HTML elements.

HTML



Structure

<html> Structure

- HTML is organized as a collection of family tree relationships.
- Child element: when an element is contained inside another element. The child element is *nested* inside of the parent element.
- Multiple levels of nesting: grandchildren, great-grandchildren, and beyond.
- Sibling element: several child element nested inside the same parent.
- The relationship between elements and their ancestor and descendent elements is known as hierarchy.
 - Indentation

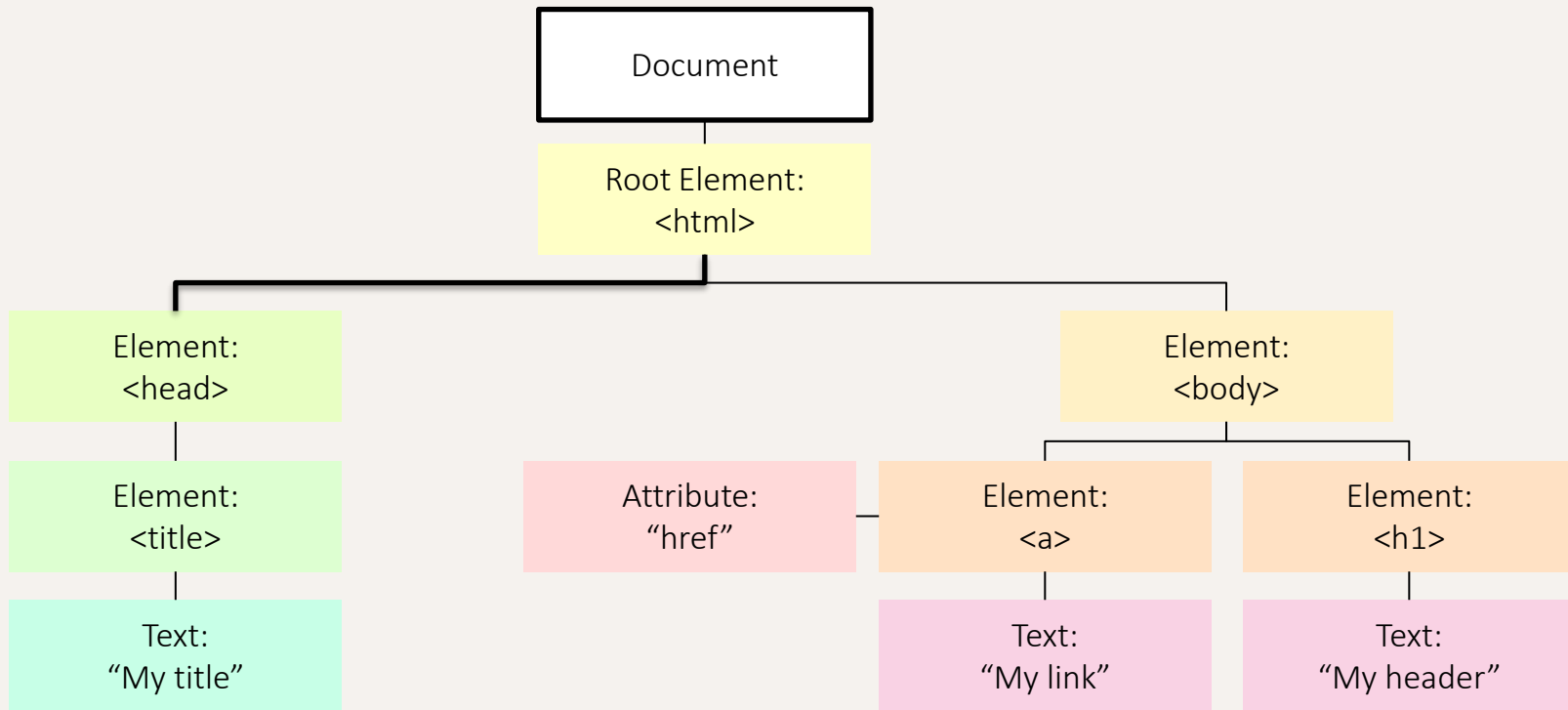
HTML Page Basic Structure

A visualization of an HTML page structure →

In VS Code- pressing !+enter provides the entire boilerplate automatically

```
<html>  
  <head>  
    <title>      content      </title>  
  </head>  
  <body>  
    <h1>      content      </h1>  
    <p>  
      content  
    </p>  
  </body>  
</html>
```

DOM- Document Object Model



HTML



Basic Elements

element

content

<p>hello world</p>

opening tag

closing tag

What is an HTML Element?

- HTML elements are defined by a start tag, content and an end tag:

`<tag name>Content goes here...</tag name>`

- The HTML element is everything from the start tag to the end tag:

`<h1>My First Heading</h1>`

`<p>My first paragraph.</p>`

`<br>` `<hr>`

Basic html Document

1. `<!DOCTYPE html>`
2. `<html>`
3. `<head>`
4. `<title>Page Title</title>`
5. `</head>`
6. `<body>`
7. `<h1>My First Heading</h1>`
8. `<h2>This is very exciting!</h2>`
9. `<p>My first paragraph.<br> <strong>Cool!</strong></p>`
10. `</body>`
11. `</html>`

My First Heading

Welcome To Web Development Course

This is a paragraph.
Cool!!

<!DOCTYPE> declaration

- Document type declaration <!DOCTYPE html> is required as the first line of an HTML document.
- WHY?
- The doctype declaration is an instruction to the browser about what type of document to expect and which version of HTML is being used, in the case of HTML5 it's:

<!DOCTYPE html>

```
1. <!DOCTYPE html>
2. <html>
3. <head>
4. <title>Page Title</title>
5. </head>
6. <body>
7. <h1>My First Heading</h1>
8. <h2>This is very exciting!</h2>
9. <p>My first paragraph.<br> <strong>Cool!</strong></p>
10.</body>
11.</html>
```

<html> element

- First element after doctype declaration opening <html> and last on page closing </html>.
- <html> element is wrapping the entire web page

```
1. <!DOCTYPE html>
2. <html>
3. <head>
4. <title>Page Title</title>
5. </head>
6. <body>
7. <h1>My First Heading</h1>
8. <h2>This is very exciting!</h2>
9. <p>My first paragraph.<br> <strong>Cool!</strong></p>
10.</body>
11.</html>
```

<head> element

- <head> element contains general information about an HTML page that isn't displayed on the page itself.
- The information in the <head> is called metadata.
- <head> includes things like:
 - i. title
 - ii. links to stylesheets
 - iii. Characters used in the document (e.g. UTF-8)

```
1. <!DOCTYPE html>
2. <html>
3. <head>
4. <title>Page Title</title>
5. </head>
6. <body>
7. <h1>My First Heading</h1>
8. <h2>This is very exciting!</h2>
9. <p>My first paragraph.<u>Cool!</u> <strong>Cool!</strong></p>
10.</body>
11.</html>
```

<title> element

- The <title> element contains a text that defines the title of an HTML document.
- The title is displayed in the browser's title bar or tab in which the HTML page is displayed.
- The <title> element can only be contained inside a document's <head> element.

```
1. <!DOCTYPE html>
2. <html>
3. <head>
4. <title>Page Title</title>
5. </head>
6. <body>
7. <h1>My First Heading</h1>
8. <h2>This is very exciting!</h2>
9. <p>My first paragraph.<br> <strong>Cool!</strong></p>
10.</body>
11.</html>
```



<title> Usage

- Required in HTML documents!
- WHY??
- The contents of a page title is very important for search engine optimization (SEO)!
- The <title> element:
 - Provides a title for the page when it is added to favorites
 - Displays a title for the page in search-engine results

<title> tips

- Include relevant keywords
- Write for the user, not for the search engine (SE)
- Try long sentences, but pay attention to length
 - Search engines will display about 50-60 characters of the title, so try not to have titles longer than that
- Be specific- try to make the title as accurate and meaningful as possible!

<body> elements

- A key HTML element used to build a webpage is the <body> element.
- Only content inside <body> tags can be displayed to the screen.
- Once the file has a <body>, many different types of content – including text, images, and buttons – can be added to the body.

```
1. <!DOCTYPE html>
2. <html>
3. <head>
4. <title>Page Title</title>
5. </head>
6. <body>
7. <h1>My First Heading</h1>
8. <h2>This is very exciting!</h2>
9. <p>My first paragraph.<u>Cool!</u> <strong>Cool!</strong></p>
10.</body>
11.</html>
```

<h1>-<h6> heading elements

- HTML can use six different levels of heading elements.
- The heading elements are ordered from the highest level <h1> to the lowest level <h6>.

```
<h1> this is h1</h1>  
<h2> this is h2</h2>  
<h3> this is h3</h3>  
<h4> this is h4</h4>  
<h5> this is h5</h5>  
<h6> this is h6</h6>
```

this is h1

this is h2

this is h3

this is h4

this is h5

this is h6

heading elements- tips

- Usually, we will use only one h1 heading in our website
- Don't jump from h1 to h3. use the various heading levels in their order.

<p> element

- The <p> element defines a paragraph.
- Browsers automatically add a single blank line before and after each <p> element.

```
<p> First Paragraph. Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.</p>  
<p> Second Paragraph. It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).</p>
```

First Paragraph. Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book. It has survived not only five centuries, but also the leap into electronic typesetting, remaining essentially unchanged. It was popularised in the 1960s with the release of Letraset sheets containing Lorem Ipsum passages, and more recently with desktop publishing software like Aldus PageMaker including versions of Lorem Ipsum.

Second Paragraph. It is a long established fact that a reader will be distracted by the readable content of a page when looking at its layout. The point of using Lorem Ipsum is that it has a more-or-less normal distribution of letters, as opposed to using 'Content here, content here', making it look like readable English. Many desktop publishing packages and web page editors now use Lorem Ipsum as their default model text, and a search for 'lorem ipsum' will uncover many web sites still in their infancy. Various versions have evolved over the years, sometimes by accident, sometimes on purpose (injected humour and the like).

**
 element**

- The
 line break element will create a line break in text and is especially useful where a division of text is required, like in a postal address or song lyrics.
- The
 element requires only an opening tag and must not have a closing tag (a void element)

<hr> element

- The <hr> element is displayed as a horizontal rule that is used to separate content (or define a change) in an HTML page.

```
<h1>William Blake </h1>
```

```
<p>17 south molton <h1>William Blake </h1>
```

```
London<br>
```

```
W1K 5QT<br>
```

```
UK<br>
```

```
</p>
```

```
<hr>
```

```
<p>William Blake (
```

```
during his life, B </p>
```

```
Age. What he called his propnetic works were said by 20th-century crit
```

```
to its merits the least read body of poetry in the English language".[2]
```

```
Jonathan Jones to proclaim him "far and away the greatest artist Britain
```

```
placed at number 38 in the BBC's poll of the 100 Greatest Britons.[4] Wh
```

```
for three years spent in Felpham,[5] he produced a diverse and symbolica
```

```
imagination as "the body of God"[6] or "human existence itself".[7]</p>
```

William Blake

17 south molton street

William Blake

17 south molton street London W1K 5QT UK

William Blake (22 November 1757 – 9 August 1827) was an English poet, painter, and printmaker. Largely unrecognised during his life, Blake is now considered a seminal figure in the history of the poetry and visual art of the Romantic Age. What he called his "prophetic works" were said by 20th-century critic Northrop Frye to form "what is in proportion to its merits the least read body of poetry in the English language".[2] His visual artistry led 21st-century critic Jonathan Jones to proclaim him "far and away the greatest artist Britain has ever produced".[3] In 2002, Blake was placed at number 38 in the BBC's poll of the 100 Greatest Britons.[4] While he lived in London his entire life, except for three years spent in Felpham,[5] he produced a diverse and symbolically rich collection of works, which embraced the imagination as "the body of God"[6] or "human existence itself".[7]

Text Formatting elements

Element	Format	HTML Code	Displayed as
	Bold text	Bold text	Bold text
	Important text	Important text	Important text
<i>	Italic text	<i> Italic text</i>	<i>Italic text</i>
	Emphasized text	 Emphasizes text	<i>Emphasizes text</i>
<mark>	Marked text	<mark> Marked text</mark>	Marked text
<small>	Smaller text	Small text	Small text
	Deleted text	 Deleted text	Deleted text
<ins>	Inserted text	<ins> Inserted text</ins>	<u>Inserted text</u>
<sub>	Subscript text	Text_{Subscripted text}	Text Subscripted text

<!--comment--> Comments

- Comments are very important in any code project.
- Comments are not displayed by the browser.
- Comments improve understanding of developers' intention while writing a specific code.
- Don't assume that another developer or even your self will later understand your intentions.

```
<!DOCTYPE html>
<html>
  <!-- Write your comments here -->
</html>

<head>
  <title>Flexbox Pricing Table</title>
  <link rel="preconnect" href="https://fonts.googleapis.com"> <!-- this is a link to a font in google fonts -->
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Sono:wght@400;700&display=swap" rel="stylesheet">
```

Class exercise 1

Programming Language for web development

Top 3 Programming Languages you need to know

HTML

This is the basic web development language used for websites content.

CSS

This language is used to style websites.

Javascript

Using Javascript, we can add functionality to our websites.

HTML



Attributes

```
<p attribute="value">hello world</p>
```

Attributes

- All HTML elements can have attributes
- Attributes provide additional information about elements
- Attributes are always specified in the start tag
- Attributes usually come in name/value pairs like: name="value"
- Example: id attribute

`<p id="Introduction">Talking about basic HTML</p>`

Attributes - Examples

- *href* in <a> specifies the URL of the page the link goes to
- *src* in specifies the path to the image to be displayed
- *width* and *height* in provide size information for images
- *alt* in provides an alternate text for an image
- *style* is used to add styles to an element, such as color, font, size, and more
- *lang* in <html> tag declares the language of the Web page

Global Attributes

- Global attributes are attributes common to all HTML elements; they can be used on all elements, though they may have no effect on some elements.

Global Attributes

Attribute	Description
accesskey	Specifies a shortcut key to activate/focus an element
class	Specifies one or more class names for an element (refers to a class in a style sheet)
contenteditable	Specifies whether the content of an element is editable or not
data-*	Used to store custom data private to the page or application
dir	Specifies the text direction for the content in an element
draggable	Specifies whether an element is draggable or not
hidden	Specifies that an element is not yet, or is no longer, relevant
id	Specifies a unique id for an element
lang	Specifies the language of the element's content
spellcheck	Specifies whether the element is to have its spelling and grammar checked or not
style	Specifies an inline CSS style for an element
tabindex	Specifies the tabbing order of an element
title	Specifies extra information about an element
translate	Specifies whether the content of an element should be translated or not

Unique *id* Attributes

- id attributes are specific and unique and can be assigned to different elements in order to differentiate between them.
- id value can be called upon by CSS and JavaScript to manipulate, format, and perform specific instructions on that element and that element only.
- Valid id attributes should begin with a letter and should only contain letters (a-Z), digits (0-9), hyphens (-), underscores (_), and periods (.).

```
<html>
<body>
  <h1 id="myHeader">Hello World!</h1> <!--this h1 was assigned with id= "me header"-->
  <button onclick="displayResult()">Change text</button>
  <script>

    function displayResult() {
      document.getElementById("myHeader").innerHTML = "Have a nice day!";}
  </script>
</body>
</html>
```

Class Attribute

- The class attribute specifies one or more class names for an element.
- The class attribute is mostly used to point to a class in a style sheet. However, it can also be used by a JavaScript (via the HTML DOM) to make changes to HTML elements with a specified class.

```
<html>
<head>
  <style>
    .intro{color: blue;}
    .important{ color:green;}
  </style>
</head>
<body>
  <h1 class="intro">Header 1</h1>
  <p>A paragraph.</p>
  <p class="important">Note that this is an important paragraph.</p>
  <p> this is a regular paragraph</p>
  <p class="important"> This is also an important paragraph.</p>
</body>
</html>
```

Style Attribute

- The style attribute is used to add styles to an element, such as color, font, size, and more

Declaration Declaration

Style= “color:blue ;font-size:12px;”

↑ ↑ ↑ ↑

Property Value Property Value

The diagram illustrates the structure of a CSS style declaration. It shows the text 'Style= “color:blue ;font-size:12px;”' with labels and arrows pointing to its components. Above the text, the word 'Declaration' is written twice, centered over the first and second parts of the style string. Below the text, four arrows point upwards to specific parts: the first arrow points to 'color' (labeled 'Property'), the second points to 'blue' (labeled 'Value'), the third points to 'font-size' (labeled 'Property'), and the fourth points to '12px' (labeled 'Value').

Styling Properties: Color

- Background Color:

```
<h1 style="background-color:Red;">Hello World</h1>
```

Hello World

- Text Color:

```
<h1 style="color:Blue;">Hello World</h1>
```

Hello World

- Border Color:

```
<h1 style="border:2px solid Tomato;">Hello World</h1>
```

Hello World

Styling Properties: Colors

- HTML colors are specified with:
- Predefined color names (red, green, blue, etc.)
- RGB value: `rgb(255, 99, 71)`
- Hex value: `#ff6347`
- HSL value: `hsl(9, 100%, 64%)`
- RGBA value: RGB + 4th value for opacity between 0 and 1
- HSLA value: HSL + 4th value for opacity between 0 and 1

HTML



More basic elements

<a> Element

- The <a> anchor element is used to create hyperlinks in an HTML document.
- hyperlinks point to other webpages, files on the same server, a location on the same page, or any other URL via the hyperlink reference attribute (href).
- The href determines the location the anchor element points to.
- It has the following syntax:

```
<a href="url">link text</a>
```

The *href* Attribute

- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.
- The link text is the part that will be visible to the reader.
- Clicking on the link text, will send the reader to the specified URL address.

The *target* Attribute

- The target attribute specifies where to open the linked document.
- By default, the linked page will be displayed in the current browser window.

```
<a href="https://www.blabla.com" target="_blank">bla-bla</a>
```

- The target attribute can have one of the following values:
 - A. `_self` - Default. Opens the link in the same window/tab as it was clicked
 - B. `_blank` - Opens the link in a new window or tab
 - C. `_parent` - Opens the link in the parent frame
 - D. `_top` - Opens the link in the full body of the window

 Element

- HTML image elements embed images in documents.
- is an empty element meaning it should not have a closing tag.
- The tag has two required attributes:
 - src – the path to the image
 - alt - Specifies an alternate text for the image

```

```

The *src* Attribute

- Specifies the path to the image
- The src attribute contains the image URL and is mandatory
- Can be relative or not
- Images with relative path must be stored in the same folder as the document

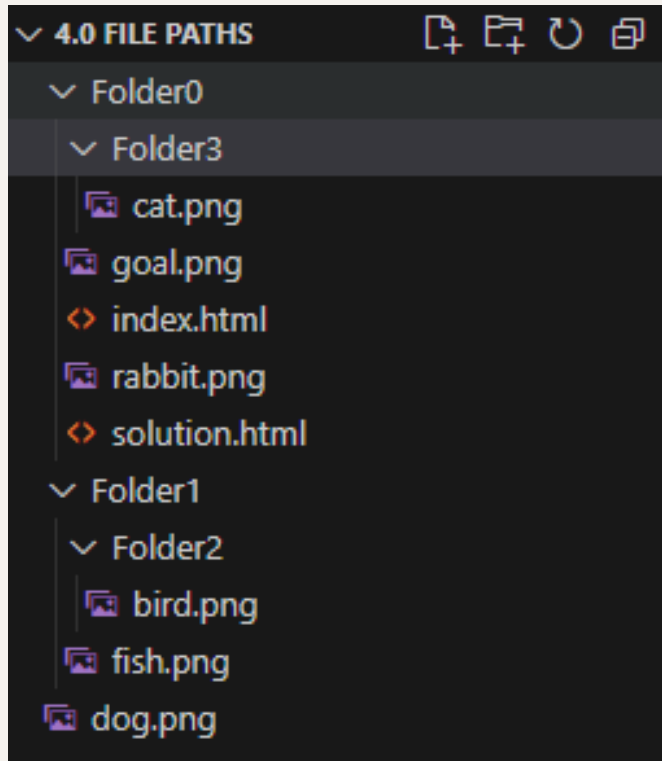
The *alt* Attribute

- An element can have alternative text via the alt attribute.
- The alternative text will be displayed if an image fails to render due to various reasons.
- The text will be read aloud if screen reading software is used and helps support visually impaired users by providing a text descriptor for the image content on a webpage.

File Path

Path	Description
<code></code>	The "picture.jpg" file is located in the same folder as the current page
<code></code>	The "picture.jpg" file is located in the images folder in the current folder
<code></code>	The "picture.jpg" file is located in the images folder at the root of the current web
<code></code>	The "picture.jpg" file is located in the folder one level up from the current folder

File Path



```
1 <h1>All the Animals</h1>
2 <h2>Rabbit:</h2>
3 
4 <h2>Cat:</h2>
5 
6 <h2>Dog:</h2>
7 
8 <h2>Fish:</h2>
9 
10 <h2>Bird:</h2>
11 
```

Recap

element

content

<p>hello world</p>

opening tag

closing tag

Basic html Document

1. `<!DOCTYPE html>`
2. `<html>`
3. `<head>`
4. `<title>Page Title</title>`
5. `</head>`
6. `<body>`
7. `<h1>My First Heading</h1>`
8. `<h2>This is very exciting!</h2>`
9. `<p>My first paragraph.<br> <strong>Cool!</strong></p>`
10. `</body>`
11. `</html>`

My First Heading

Welcome To Web Development Course

This is a paragraph.
Cool!!

`<div>` Element

- One of the most common elements in HTML is the `<div>` element.
- `<div>` - short for “division” – is a container that divides the page into sections.
- These sections are very useful for grouping elements in your HTML together.
- Any sort of content can be put inside the `<div>` tag!
- easily styled by using the class or id attribute.

 Element

- tag is an inline container used to mark up a part of a text, or a part of a document.
- The tag is easily styled by CSS or manipulated with JavaScript using the class or id attribute.
- The tag is much like the <div> element, but <div> is a block-level element and is an inline element.

<div> element 1

<div> element 2

<div> element 3

 element 1 element 2 element 3

Element Display

- The display property is the most important property for controlling layout.
- Every HTML element has a default display value depending on what type of element it is.
- Basic default display values: inline and block.
- A block-level element always starts on a new line and takes up the full width available.
- An inline element does not start on a new line, and it only takes up as much width as necessary.

Block-Level Elements

- `<address>`
- `<article>`
- `<aside>`
- `<blockquote>`
- `<canvas>`
- `<dd>`
- `<div>`
- `<dl>`
- `<dt>`
- `<fieldset>`
- `<figcaption>`
- `<figure>`
- `<footer>`
- `<form>`
- `<h1>-<h6>`
- `<header>`
- `<hr>`
- `<li>`
- `<main>`
- `<nav>`
- `<noscript>`
- `<ol>`
- `<p>`
- `<pre>`
- `<section>`
- `<table>`
- `<tfoot>`
- `<ul>`
- `<video>`

Inline Elements

- `<a>`
- `<abbr>`
- `<acronym>`
- `<b>`
- `<bdo>`
- `<big>`
- `<br>`
- `<button>`
- `<cite>`
- `<code>`
- `<dfn>`
- `<em>`
- `<i>`
- `<img>`
- `<input>`
- `<kbd>`
- `<label>`
- `<map>`
- `<object>`
- `<output>`
- `<q>`
- `<samp>`
- `<script>`
- `<select>`
- `<small>`
- `<span>`
- `<strong>`
- `<sub>`
- `<sup>`
- `<textarea>`
- `<time>`
- `<tt>`
- `<var>`

The Display property: inline-block Value

- Compared to **display: inline**, the major difference is that **display: inline-block** allows to set a width and height on the element.
- Compared to **display: block**, the major difference is that **display: inline-block** does not add a line-break after the element, so the element can sit next to other elements.

HTML



Lists

`<ul>` / `<ol>` HTML list element

`<ul>` An unordered HTML list:

- Item
- Item
- Item
- Item

`<ol>` An ordered HTML list:

1. First item
2. Second item
3. Third item
4. Fourth item

 List Item Element

- The list item element create list items inside:
 - I. Ordered lists
 - II. Unordered lists
- Listing items with element

Unordered list element

```
<ul>  
  <li>gin bombay sapphire</li>  
  <li>tonic water</li>  
  <li>squeeze of lemon</li>  
  <li>ice</li>  
</ul>
```

Unordered list:

- gin bombay sapphire
- tonic water
- squeeze of lemon
- ice

Ordered list element

```
<ol>  
  <li>gin bombay sapphire</li>  
  <li>tonic water</li>  
  <li>squeeze of lemon</li>  
  <li>ice</li>  
</ol>
```

Orederd list:

1. gin bombay sapphire
2. tonic water
3. squeeze of lemon
4. ice

Description Lists

```
<dl>
  <dt>Coffee</dt>
  <dd>- black hot drink</dd>
  <dt>Milk</dt>
  <dd>- white cold drink</dd>
</dl>
```

Description list list:

Coffee
- black hot drink

Milk
- white cold drink

List Elements

Tag	Description
	Defines an unordered list
	Defines an ordered list
	Defines a list item
<dl>	Defines a description list
<dt>	Defines a term in a description list
<dd>	Describes the term in a description list

Class exercise 2

It's My Birthday!

On the 29th September



What to bring?

- Ballons
- Cake
- Present

[Link to google maps](#)

HTML



Tables

<table> element

- In HTML, the <table> element has content that is used to represent a two-dimensional table made of rows and columns.

<table>

All table content

</table>

<tr> element

- The table row element, <tr>, is used to add rows to a table before adding table data and table headings.

```
<table>
  <tr>
    content of first row
  </tr>
  <tr>
    content of second row
  </tr>
</table>
```

<td> element

- The table data element, <td>, can be nested inside a table row element, <tr>, to add a cell of data to a table.

```
<table>
  <tr>
    <td>content-A</td>
    <td>content-B</td>
  </tr>
  <tr>
    <td>content-C</td>
    <td>content-D</td>
  </tr>
</table>
```

content-A	content-B
content-C	content-D

<th> element

- The table heading element, <th>, is used to add titles to rows and columns of a table and must be enclosed in a table row element, <tr>.

```
<table>
  <tr>
    <th>content-A</th>
    <th>content-B</th>
  </tr>
  <tr>
    <td>content-C</td>
    <td>content-D</td>
  </tr>
</table>
```

content-A	content-B
content-C	content-D

colspan attribute

- The colspan attribute on a table header <th> or table data <td> element indicates how many columns that particular cell should span within the table. The colspan value is set to 1 by default and will take any positive integer between 1 and 1000.

```
<table>
  <tr>
    <th colspan="2">content-A</th>
    <th>content-B</th>
  </tr>
  <tr>
    <td>content-C</td>
    <td>content-D</td>
  </tr>
</table>
```

content-A		content-B
content-C	content-D	

rowspan attribute

- Similar to colspan, the rowspan attribute on a table header or table data element indicates how many rows that particular cell should span within the table. The rowspan value is set to 1 by default and will take any positive integer up to 65534.

HTML

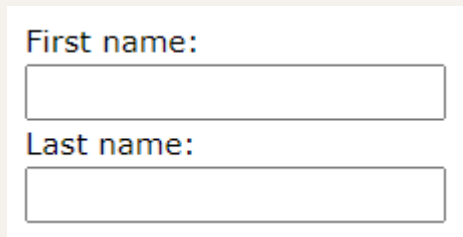


Forms

<form> Element

- An HTML <form> tag is used to collect user input. The user input is most often sent to a server for processing.
- Example-

```
<form>  
  <label for="fname">First name:</label><br>  
  <input type="text" id="fname" name="fname"><br>  
  <label for="lname">Last name:</label><br>  
  <input type="text" id="lname" name="lname">  
</form>
```



First name:

Last name:

`<input>` Element

- The `<input>` element is the most important form element.
- An `<input>` element can be displayed in many ways, depending on the type attribute.

<input> Attributes: Type

- button
- checkbox
- color
- date
- datetime-local
- email
- file
- hidden
- image
- month
- number
- password
- radio
- range
- reset
- search
- submit
- tel
- text
- time
- url
- week

<input> Attributes: Name, ID

- 2 attributes that aim to support data transfer
- ID will serve for manipulations
- Name will serve to send the data to the required location with the name value attached.
- Should match each other (i.e., same value is possible)
- Should also match the “for” attribute of label element.
- Each input field must have a name attribute to be submitted.
- **Note: Only input elements with a name attribute will have their values passed when submitting a form.**

<input> Attributes

Attribute name	Value type	Description
form	Form id	Specifies the form the <input> element belongs to
name	text	Specifies the name of an <input> element
Required	required	Specifies that an input field must be filled out before submitting the form
Value	text	Specifies the default value of an <input> element
Autocomplete	on off	Specifies whether an <input> element should have autocomplete enabled- based on previous values entered
checked	checked	Specifies that an <input> element should be pre-selected when the page loads (for checkboxes, radio buttons, and list boxes)

<label> Element

- The <label> tag defines a label for many form elements.
- The <label> element is useful for screen-reader users, because the screen-reader will read out loud the label when the user is focused on the input element.
- The <label> element also help users who have difficulty clicking on very small regions (such as radio buttons or checkboxes) - because when the user clicks the text within the <label> element, it toggles the radio button/checkbox.
- **The “for” attribute of the <label> tag should be equal to the id attribute of the <input> element to bind them together.**

<label> Attributes

Attribute	Value	Description
for	element_id	Specifies the id of the input element the label should be bound to
form	form_id	Specifies which form the label belongs to

<button> Element

- The <button> HTML element is an interactive element activated by a user with a mouse, keyboard, finger, voice command, or other assistive technology.
- Once activated, it then performs a programmable action, such as submitting a form or opening a dialog.

<button> Attributes

Attribute	Value	Description
Autofocus	autofocus	Specifies that a button should automatically get focus when the page loads
Disabled	disabled	Specifies that a button should be disabled
Form	form_id	Specifies which form the button belongs to
name	name	Specifies a name for the button
type	Button- default Reset Submit	Specifies the type of button Reset- resets the fields in the form Submit- submits the form
value	text	Specifies an initial value for the button

<input> Submit Button Element

- The `<input type="submit">` defines a button for submitting the form data to a form-handler.
- The form-handler is typically a file on the server with a script for processing input data.
- The form-handler is specified in the form's action attribute.

<form> Data Destination

- When a user submits the form, information in these input elements is passed to the source which is named in the action attribute of the form.

<Form> Attributes

Attribute	Value	Description
action	URL	Specifies where to send the form-data when a form is submitted
method	get post	Specifies the HTTP method to use when sending form-data
name	text	Specifies the name of a form

Form Method Attribute

- The method attribute specifies the HTTP method (GET or POST) to be used when submitting the form data.
- When to Use GET?
- The default HTTP method when submitting form data is GET.
- However, when GET is used, the form data will be visible in the page's address field.
- When to Use POST?
- Always use POST if the form data contains sensitive or personal information. The POST method does not display the form data in the page address field.

HTML



Inspect HTML

How to View HTML Source?

- **View HTML Source Code:**

Right-click in an HTML page and select "View Page Source" (in Chrome), or similar in other browsers.

- **Inspect an HTML Element:**

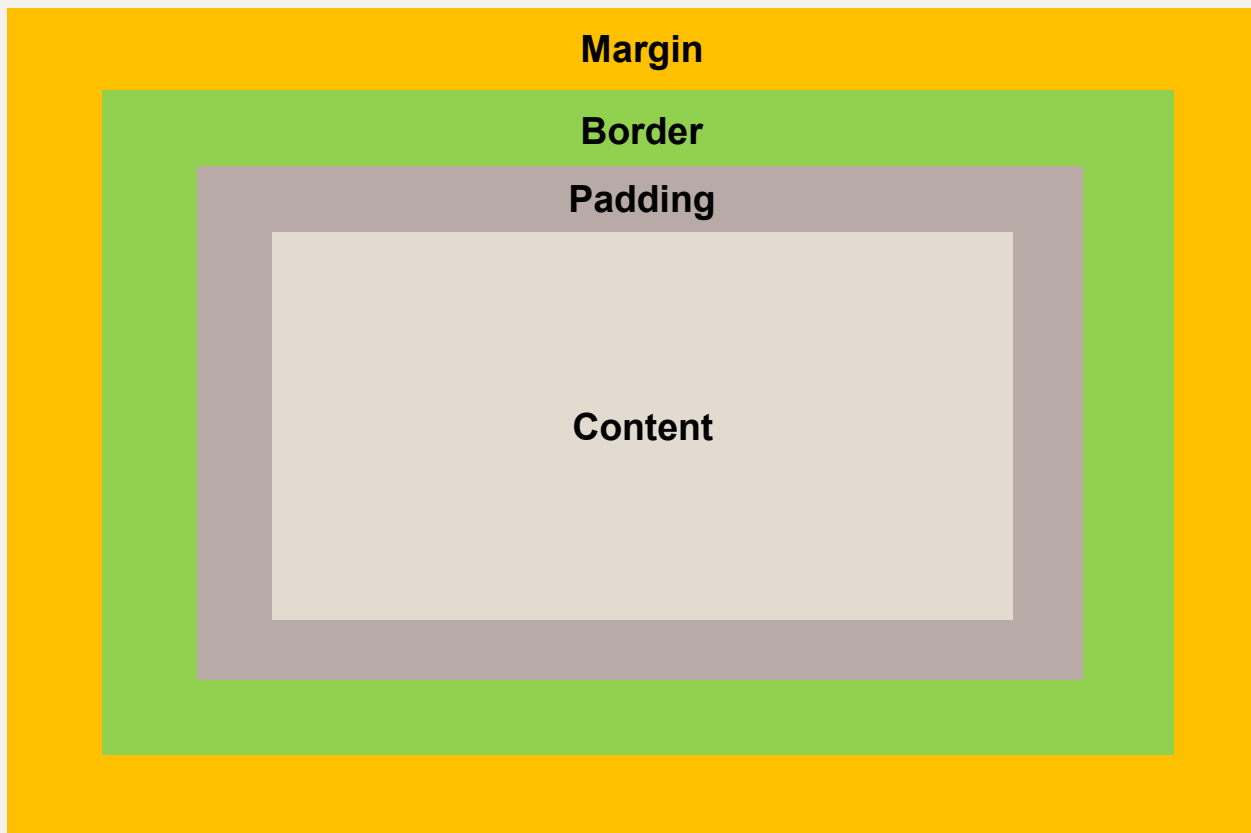
Right-click on an element (or a blank area), and choose "Inspect" or "Inspect Element" to see what elements are made up of (you will see both the HTML and the CSS). You can also edit the HTML or CSS on-the-fly in the Elements or Styles panel that opens.

HTML

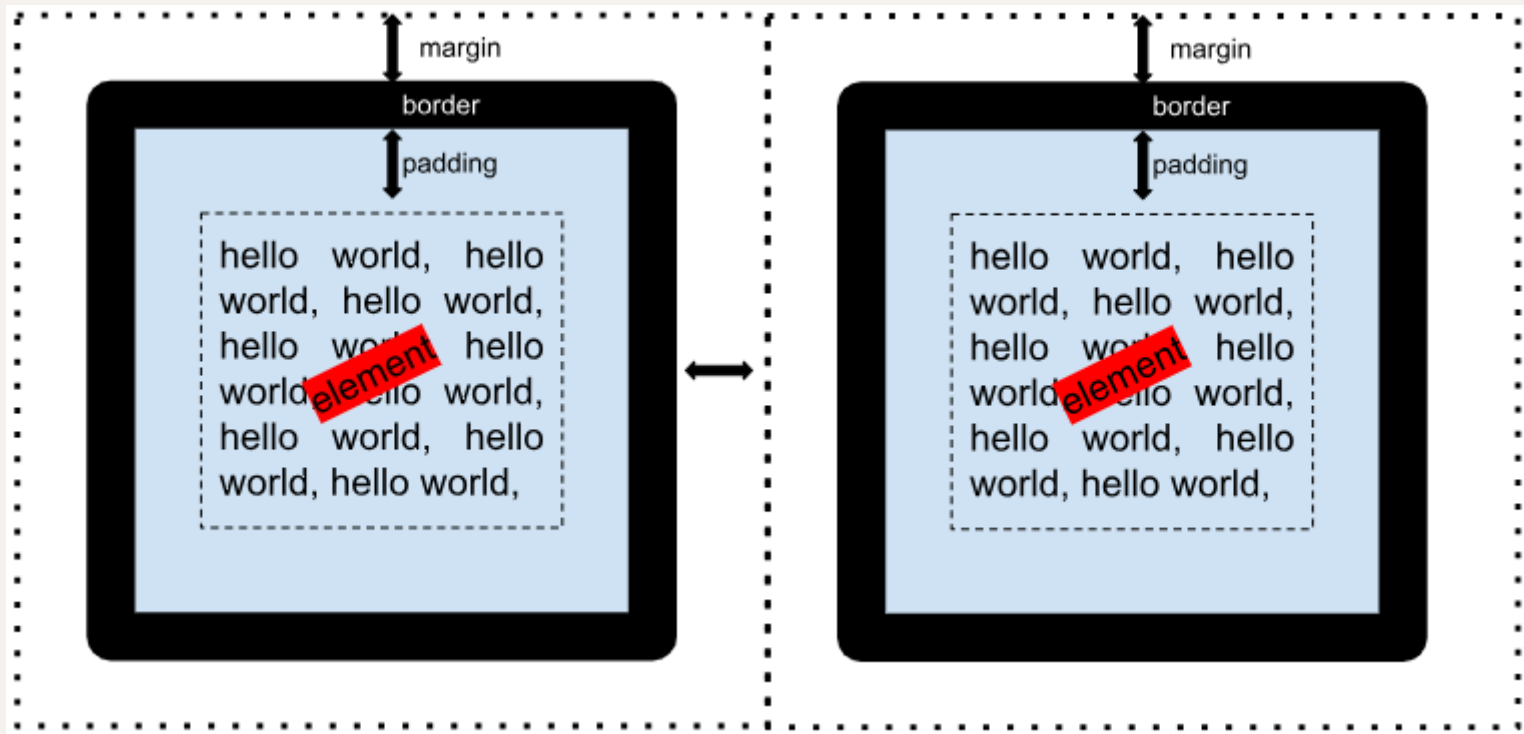


Box Model

Box Model

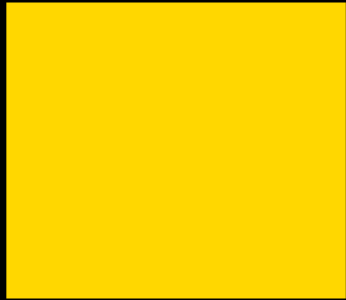


Box Model



Box Model- Div's Example

Lorem ipsum dolor sit amet,
consectetur adipiscing elit.
Aliquam at sapien porttitor
urna elementum lacinia. In id
magna pulvinar, ultricies
lorem id, vehicula elit.
Aliquam eu luctus nisl, vitae
pellentesque magna. Phasellus
dolor metus, laoreet ac
convallis sit amet, efficitur sed
dolor.



Margin Property

- The outer layer that creates space around elements, outside of any defined borders.
- Use general margin property
- or separately:
 - I. margin-top
 - II. margin-right
 - III. margin-bottom
 - IV. margin-left

```
style="margin: 50px;"
```

Border Property

- Border is a style property with a 3 parameters value:
 - I. Width
 - II. Style
 - III. Color
- The border property is a shorthand property for 3 border properties- In the following syntax:

```
Style="border: 5px solid red"
```

- The border property is also a shorthand property for all separate 4 borders properties:
 - I. Border-top
 - II. Border-right
 - III. Border-bottom
 - IV. Border-left

Padding Property

- Padding creates extra space within an element, while margin creates extra space around an element.
- The padding property is a shorthand property for:
 - I. padding-top
 - II. padding-right
 - III. padding-bottom
 - IV. padding-left

```
Style="padding: 50px;"
```